

Requirements–Design Traceability

1.1 Requirements–Design Traceability Table

This section presents the Requirements–Design Traceability Table (RDT), which maps key functional requirements defined in the Software Requirements Specification (SRS) to corresponding design artifacts.

1.1.1 Legend

- **UC** — Use Case Diagram
- **DFD** — Data Flow Diagram
- **SD** — Sequence Diagram
- **CD** — Class Diagram
- **UI** — User Interface Prototype
- **ARCH** — System Architecture Diagram

1.1.2 User Authentication and Authorization

Req ID	Requirement Description	UC	DFD	SD	CD	UI	ARCH	Design Contents
FR-001	User Registration	UC-01	DFD-0	SD-01	CD-01	UI-01	ARCH-01	User, AuthController, UserDB, EmailService
FR-002	Admin Account Approval	UC-01	DFD-1	SD-01	CD-01	UI-02	ARCH-01	Admin, UserMgmt, NotificationService
FR-003	Player Login	UC-01	DFD-0	SD-02	CD-01	UI-03	ARCH-01	LoginForm, AuthService, SessionManager
FR-004	Admin Login	UC-01	DFD-0	SD-02	CD-01	UI-04	ARCH-01	AdminLogin, AuthService, RBAC Module
FR-005	Password Reset	UC-01	DFD-1	SD-03	CD-01	UI-05	ARCH-01	ResetForm, TokenService, EmailService
FR-006	Session Management	UC-01	DFD-0	SD-02	CD-01	UI-03	ARCH-01	SessionManager, Timeout, Logger

1.1.3 Player Module and Learning Features

Req ID	Requirement Description	UC	DFD	SD	CD	UI	ARCH	Design Contents
FR-007	Player Dashboard	UC-01	DFD-0	–	CD-01	UI-06	ARCH-01	Dashboard, StatsCard, RecentActivity
FR-010	Simulation Viewing	UC-01	DFD-2	SD-04	CD-01	UI-09	ARCH-01	SimulationPlayer, AnimationEngine
FR-011	Performance Tracking	UC-01	DFD-2	SD-04	CD-01	UI-06	ARCH-01	Tracker, AnalyticsService, LogDB
FR-013	Help Desk Ticket	UC-01	DFD-1	SD-05	CD-01	UI-11	ARCH-01	TicketForm, HelpDeskService
FR-015	Event Viewing	UC-01	DFD-1	SD-07	CD-01	UI-13	ARCH-01	EventCalendar, AnnouncementCard

1.1.4 Administration, Content, and Events

Req ID	Requirement Description	UC	DFD	SD	CD	UI	ARCH	Design Contents
FR-016	Admin Dashboard	UC-01	DFD-0	ACT-02	CD-01	UI-14	ARCH-01	AdminDashboard, SystemMetrics
FR-019	Upload Simulation	UC-01	DFD-2	SD-09	CD-01	UI-17	ARCH-01	ContentUpload, FileValidator, Media
FR-021	Help Desk Management	UC-01	DFD-2	SD-06	CD-01	UI-19	ARCH-01	TicketManager, ResponseInterface
FR-022	Event Management	UC-01	DFD-2	SD-07	CD-01	UI-20	ARCH-01	EventCreator, Scheduler, NotifService
FR-023	Analytics/Reporting	UC-01	DFD-2	ACT-03	CD-01	UI-21	ARCH-01	ReportEngine, DataExporter, Visuals

1.2 Design Assumptions and Constraints

1.2.1 Design Assumptions

1. Users have basic web browsing skills and cricket knowledge.
2. Admin approval will occur within 24-48 hours of registration.
3. University network provides adequate bandwidth for media streaming.

1.2.2 Design Constraints

1. **Technology:** Limited to open-source frameworks.
2. **Budget:** Zero-budget project.
3. **Timeline:** Project must be completed within 1 academic year.

1.3 Key Design Decisions

1.3.1 Decomposition of Processes in DFDs

The decision to decompose processes across three DFD levels was made to achieve clarity and maintainability. Level 0 defines the boundary, while Level 2 provides logic for complex simulation features.

1.3.2 Selection of Class Relationships

Relationships were selected based on domain analysis:

- **Inheritance:** Player and Admin extend the User class.
- **Composition:** Simulation contains Steps (strict lifecycle).
- **Association:** Link between User and Ticket.

1.3.3 Distribution of Functionality Across Multiple Diagrams

Functionality was distributed to ensure separation of concerns. Sequence Diagrams handle timing, while Class Diagrams handle structure.

1.3.4 Other Key Design Decisions

1. **Layered Architecture:** Chose a 3-tier structure for modularity.
2. **RBAC:** Implemented for secure separation of roles.
3. **Responsive UI:** Ensuring usability on various screen sizes.